

Roll no: 26k-0006

## Ques no 4 (Online Shopping Bill Calculator)

### Algorithm:

- 1) Start
- 2) Read Quantity, Price-Per-Item, Discount-Percentage and Tax-Percentage.
- 3) IF Quantity  $\leq 0$  OR Price-Per-Item  $\leq 0$  OR Discount-Percentage  $< 0$  OR Tax-Percentage  $< 0$   
Display "Error: Invalid input values, Calculation Terminated."
- 4) Calculate Subtotal  $(s) = \text{Quantity}(q) \times \text{Price-Per-Item}(p)$
- 5) Calculate Discount Amount  $(d) = \frac{\text{Subtotal} \times \text{Discount-Percentage}(d\text{-pct})}{100}$
- 6) Calculate Final Bill (a)  $\text{Final Bill} = \text{Subtotal} - \text{Discounted-Amount} + \frac{\text{Discounted-Amount} \times \text{Tax-Percentage}(t)}{100}$
- 7) Display subtotal, Discount Applied, Tax Added.
- 8) Stop

### PAC:

| Given Data                                                                                                                                                                                                                    | Processing Required                                                                                                                                                                                                                                                                                | Output Requirement                                                                                                                 |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li><math>q</math> (quantity of products)</li><li><math>p</math> (Price-per-Item)</li><li><math>d\text{-pct}</math> (Discount-percentage)</li><li><math>t</math> (Tax-Percentage)</li></ul> | <ul style="list-style-type: none"><li>Check if inputs are negative or illogical.</li><li>Formulas:<ul style="list-style-type: none"><li><math>s = q \times p</math></li><li><math>d = (s \times d\text{-pct}) / 100</math></li><li><math>a = s - d + (d \times t) / 100</math></li></ul></li></ul> | <ul style="list-style-type: none"><li>Error message if input fails</li><li>Generated subtotal and calculated final Bill.</li></ul> |

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### IPO chart:

| Input                     | Processing                                | Output                                     |
|---------------------------|-------------------------------------------|--------------------------------------------|
| • Product quantity (q)    | • Accept Transactional order values.      | • Warning Alert on invalid input.          |
| • Retail Unit Price (p)   | • Evaluate numerical constraints.         | • Structural receipt.                      |
| • Discount % rate (d-pct) | • Compute subtotal (s).                   | • Total calculated billing outcome Amount. |
| • Sales Tax % rate (t)    | • Apply ratio multiplier for discount (d) |                                            |
|                           | • Compute tax.                            |                                            |

### Pseudo code:

BEGIN

DECLARE Quantity AS INTEGER

DECLARE Price, Discount-Pct, Tax-Pct, Subtotal, Discount-Amount, Final Bill AS REAL.

PRINT "Enter Product Quantity: "

INPUT Quantity

PRINT "Enter individual price per unit item: "

INPUT Price

PRINT "Enter active discount percentage rate: "

INPUT Discount-Pct

PRINT "Enter localized sales tax percentage rate: "

INPUT Tax-Pct.

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IF (Quantity  $\leq 0$ ) OR (Price  $\leq 0$ ) OR (Discount-Pct  $\leq 0$ ) OR (Tax-Pct  $\leq 0$ ) THEN

PRINT "Error: Invalid input, Program Terminated"

TERMINATE

ENDIF

Subtotal = Quantity \* Price

Discount-Amount = (Subtotal \* Discount-Pct) / 100.0

Final-Bill = Subtotal - Discount-Amount +  $\frac{\text{Discount-Amount} * \text{Tax-Pct}}{100.0}$

PRINT "-----CUSTOMER RECEIPT-----"

PRINT "Gross Transaction Subtotal: Rs", Subtotal

PRINT "Total Price Reduction Discount: Rs", Discount-Amount

PRINT "Final Calculated Net Bill Invoice: Rs", Final-Bill

PRINT "-----"

END

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## Flowchart:

